How disrupted spatial fluxes reshape food web structure along the river–sea continuum

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Abstract

Anthropic disturbances and ecosystem processes transcend the artificial boundaries between land, freshwater, and sea. While these systems are interconnected through species and material fluxes, they are often studied in isolation. In response, the concept of metaecosystem has emerged as a powerful framework to understand how fluxes among ecosystems shape their functions and dynamics. Disrupting these fluxes can have dramatic consequences that ripple across entire landscapes. In aquatic systems, dams and nutrient pollution are known to strongly affect fish communities. Yet, little is still known about how these effects propagate and potentially accumulate, through hydrographic basins and ultimately to the sea. Using bayesian spatio-temporal statistical models (R-INLA), we aim to analyse existing data on food-web structure across France, combined with environmental variables (dams, nutrients, …), to quantify how breaking or increasing fluxes among systems disturbs food-web structure locally and how these disturbances spread throughout the metaecosystem.

Keywords: *foodweb ecology, stream network, spatial autocorrelation, bayesian statistics.*

Introduction

This is some text.

Multiple stressors

- Synergistic and antagonistic effects of multiple stressors (Jackson, Pawar, and Woodward 2021; Orr et al. 2024).
- Temperature can interact with nutrients and impact food web structure in a nonlinear way Binzer et al. (2012).
- (Danet et al. 2021).

Connectivity

- Small dispersal stabilizes through spatial structure and asynchrony (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005; Pillai, Gonzalez, and Loreau 2011).
- Prey switching of mobile top predators can stabilize connected food webs through 'spatial coupling' (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005; Pillai, Gonzalez, and Loreau 2011).
- At high dispersal/connectivity the stabilizing effect wanes as strongly connected patches behave as a single patch (LeCraw, Kratina, and Srivastava 2014; McCann, Rasmussen, and Umbanhowar 2005).
- Studies between pairs of the following connectivity, complexity and stability are common. However, studies investigating how these are interconnected are rare (LeCraw, Kratina, and Srivastava 2014).
- Connectivity has 4 dimensions: longitudinal, lateral, vertical and temporal (Grill et al. 2019).
- We identified five pressure factors to represent the main human interferences within the four dimensions of fluvial or river connectivity (Grill et al. 2019):
    (1) river fragmentation (longitudinal);
    (2) flow regulation (lateral and temporal);
    (3) sediment trapping (longitudinal, lateral and vertical);
    (4) water consumption (lateral, vertical and temporal);
    (5) infrastructure development in riparian areas and floodplains (lateral and longitudinal).
- In freshwater system, fish dispersal is limited. A meta-analysis found that the mean dispersal kernel is around 10km (Comte and Olden 2018)
- (Doretto, Piano, and Larson 2020).

Dams

- Decrease river connectivity / reduce migration (Dean et al. 2023; He et al. 2024; Pedersen et al. 2012).
- Reservoir, lotic from lentic conditions, can result in hypoxic conditions. Favor phytoplancktons and macrophytes (Ding et al. 2026; Dean et al. 2023).
- Reservoir increase residence time of nutrients upstream and decrease the downstream output, which can result in a decrease in primary productivity.
- Hydropeaking - that is, the sudden release of large water to answer the higher electric demand - can harm riverine community due to sudden change in water level and water flow.
- Cumulative impact of dams can reduce the nutrient input in deltas (Dean et al. 2023).

Methods

Food web inference

Infer missing fish lengths

Most length of fishes are measured but not all. Yet, we need individual fish length to determine their diet based on their ontogeny. We infer missing fish size using information on the given fishing operation. When there are more than three observations of the fish species, we use a truncated normal distribution (package truncdist; Nadarajah and Kotz (2006)). The mean and variance are given by the modes of the observed distribution of the fishing operation. The minimum and maximum of the distribution are given by the minimal and maximal length observed on the entire data set for that species. We do so to avoid infering irrealistic lengths.

Get fish diet

The metaweb

Subsampling the metaweb to get local food webs

Results

Discussion

Appendix

Model DAG

Put model DAG here using targets.

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